

INTEGRATED SECURITY ASSESSMENT AND FORENSICS

Project KAVACH

6

PCAP SOURCES

63

DOCUMENTED IOCS

7

WEB FINDINGS

3

WORKSTREAMS

Connecting network evidence, web vulnerabilities and business risk

IIT Roorkee x FutureNSE Capstone

ROLE FOCUS

Shamanth R Singh

Network Forensics Lead

PCAP triage, C2 identification, IOC extraction and architecture recommendations

PROJECT OBJECTIVE

Investigate a realistic multi-stage breach scenario for the fictional Meridian FinServe enterprise, then connect network-forensics evidence and web-application weaknesses into a joint threat model and executive security narrative.

EVIDENCE PRINCIPLE

Separate confirmed packet artifacts from inferred attacker objectives, document confidence, and avoid overstating what the evidence proves.

THREE-WORKSTREAM MODEL

A - NETWORK FORENSICS

PCAP triage, C2 patterns, malware staging, IOC extraction

C - STRATEGIC SYNTHESIS

Joint threat model, attack narrative, defence-in-depth

B - WEB SECURITY

DVWA, Juice Shop, OWASP findings

Evidence from both technical streams converges into one defensible business story

From packet artifacts to a unified attack hypothesis

The synthesis links separate evidence streams; it does not claim one capture proved the complete end-to-end chain.

WORKSTREAM A - NETWORK FORENSICS

1,689

C2-LIKE /ACTIVITY GETS

~5.8 MB

DIRECT-IP OUTBOUND UPLOAD

1,030

TCP NULL PROBES

1,000

UNIQUE SCANNED PORTS

- Hancitor, Trickbot and RedLine traffic supported malware staging, C2 and exfiltration-like behaviour.
- Plaintext FTP exposed valid Administrator credentials and file movement directly in packet data.

WORKSTREAM B - WEB APPLICATION SECURITY

CRITICAL	Command injection
HIGH	SQL injection
HIGH	Cross-site scripting
HIGH	IDOR / broken access control
HIGH	Unrestricted file upload
MEDIUM	Authentication weaknesses
MEDIUM	Version and dependency risk

Coverage: 7 findings across 5 OWASP Top 10 categories

POTENTIAL UNIFIED NARRATIVE - MEDIUM TO HIGH CONFIDENCE

SQL injection

Initial foothold

IDOR

Data discovery

Sensitive records

Collection

Outbound POST

Exfiltration pattern

C2 + DNS

Network evidence

Break the chain with evidence-mapped controls

Recommendations span identity, perimeter, segmentation, applications, data, observability and response.

DEFENCE-IN-DEPTH ROADMAP

Identity M MFA + session hardening	Perimeter M WAF + deny-by-default egress
Segmentation H Restrict SMB / LDAP / Kerberos	Application M Parameterized queries + authorization
Data M Encrypt + restrict record access	Observability M Correlate DNS, HTTP and app logs

EVIDENCE DISCIPLINE

CONFIRMED

Plaintext FTP credentials, file transfer and the 1,030-packet NULL scan.

STRONGLY SUPPORTED

Hancitor and Trickbot C2 patterns; RedLine high-volume direct-IP upload.

INFERRED / REQUIRES CORRELATION

Exact stolen contents and a single end-to-end event linking all workstreams.

MY CONTRIBUTION - NETWORK FORENSICS LEAD

- Performed baseline-first PCAP triage across malware, exfiltration, FTP and scanning captures.
- Identified beaconing, suspicious DNS, payload staging, lateral-movement risk and upload patterns.
- Extracted 63 IOCs, reconstructed malware-object hashes and documented confidence levels.
- Translated packet evidence into containment priorities and hardened architecture recommendations.

PORTFOLIO EVIDENCE

Explore the Project KAVACH repository

PCAPs, IOC register, web findings, threat model, defence plan, executive readout and reflections.

VIEW REPOSITORY

Docker web lab
reproducible in <15 min